

H α Transmission Spectra in TPM.py:

Ly α Pumping (Christie et al. 2013) and the $n = 2$ Population

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1. Overview

TPM.py (the Transmission Probability Model shipped with ATES) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2³S triplet (10830 Å) and the HI / D Ly α doublet (1215.67 Å).

This memo documents the addition of the H α (6562.8 Å, $n = 2 \rightarrow n = 3$) transmission spectrum. Unlike Ly α , which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, H α absorbs out of the $n = 2$ level, whose population must be computed from a non-LTE balance. We follow **Christie, Arras & Li (2013, ApJ 772, 144)**, including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and **Ly α radiative pumping** ($1s \leftrightarrow 2p$). The Ly α mean intensity $J_{\text{Ly}\alpha}(r)$ is *not* produced by ATES; it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017) — see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether `Jlya_file` is set. The existing He/Ly α behavior is unchanged.

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The 2s and 2p states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.3 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.26 \text{ s}^{-1}$). With $n_{1s} = n_{\text{HI}}$, n_e and T taken from the ATES profiles, the rate-equilibrium equations are (Christie et al. Eqs. 12–13)

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + (C_{2p \rightarrow 1s} + C_{2p \rightarrow 2s})n_e + \Gamma_{2p}] n_{2p} - C_{2s \rightarrow 2p} n_e n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e^2, \quad (1)$$

$$[(C_{2s \rightarrow 1s} + C_{2s \rightarrow 2p})n_e + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - C_{2p \rightarrow 2s} n_e n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e^2. \quad (2)$$

This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -C_{2s \rightarrow 2p} n_e \\ -C_{2p \rightarrow 2s} n_e & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The total $n = 2$ density driving H α absorption is $n_2 = n_{2s} + n_{2p}$; in practice it is dominated by the metastable $2s$ state.

2.1 Rate coefficients

The forward collisional rates are taken from Christie et al. Table 2 (Janev et al. 2003); the reverse (super-elastic) rates follow by detailed balance using the statistical weights $g_{1s} = 2$, $g_{2s} = 2$, $g_{2p} = 6$. With $t_4 \equiv T/10^4 \text{ K}$:

Process	Coefficient	Expression [$\text{cm}^3 \text{ s}^{-1}$]
Case-B recomb. (Draine 2011)	α_B	$2.54 \times 10^{-13} t_4^{-0.8163-0.0208 \ln t_4}$
$\rightarrow 2s$ (incl. cascade)	α_{2s}	$(0.282 + 0.047 t_4 - 0.006 t_4^2) \alpha_B$
$\rightarrow 2p$	α_{2p}	$\alpha_B - \alpha_{2s}$
$1s \rightarrow 2s$ excitation	$C_{1s \rightarrow 2s}$	$1.21 \times 10^{-8} t_4^{-0.455} e^{-118400/T}$
$1s \rightarrow 2p$ excitation	$C_{1s \rightarrow 2p}$	$1.71 \times 10^{-8} t_4^{-0.077} e^{-118400/T}$
$2s \leftrightarrow 2p$ ℓ -mix	$C_{2s \rightarrow 2p}$	$6.21 \times 10^{-5} (\ln(T/T_0) - \gamma) / \sqrt{T}$
$2s \rightarrow 1s$ (super-elastic)	$C_{2s \rightarrow 1s}$	$1.21 \times 10^{-8} t_4^{-0.455} (g_{1s}/g_{2s})$
$2p \rightarrow 1s$ (super-elastic)	$C_{2p \rightarrow 1s}$	$1.71 \times 10^{-8} t_4^{-0.077} (g_{1s}/g_{2p})$
$2p \rightarrow 2s$ ℓ -mix	$C_{2p \rightarrow 2s}$	$C_{2s \rightarrow 2p} (g_{2s}/g_{2p})$

with $T_0 = 1.02 \text{ K}$ and $\gamma = 0.57721$ (Euler–Mascheroni). The reverse rates are written in their analytically-cancelled form: the Boltzmann factor $e^{+118400/T}$ implied by detailed balance cancels the $e^{-118400/T}$ inside $C_{1s \rightarrow 2s}/C_{1s \rightarrow 2p}$, which avoids a numerical $0 \times \infty$ at the very low temperatures that can occur at the base of the post-processed profile.

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p}) B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s^{-1} .

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If `Jlya_file` points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1	r/R_p	(same radial coordinate as the ATES output)
column 2	$J_{\text{Ly}\alpha}$	[erg s ⁻¹ cm ⁻² Hz ⁻¹ sr ⁻¹]

linearly interpolated onto the ATES grid and clamped to the endpoints. A template ships as `inputdata/Jlya.txt.example`.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere (2.6×10^4 erg cm⁻² s⁻¹ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, TPM.py computes the absorbed LyC flux from the ATES inputs:

$$F_{\text{LyC}} = \xi \underbrace{\frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{HI}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6$ eV) and the orbital distance a are read from `input.inp`, N_{HI} is the vertical neutral-hydrogen column from the ATES profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18}$ cm². The factor ξ is the day-night / 2D flux dilution, taken to match ATES’s “2D approximate method” (Rate/2 $\rightarrow$ 0.5, Rate/4 $\rightarrow$ 0.25, else 1; read from `input.inp`, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2$ erg cm⁻² s⁻¹ for the HD 209458b input). A fixed value can be forced via `F_LyC_override` and the dilution via `xi_override`.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply `Jlya_file`.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4$ eV, $\lambda < 3647$ Å); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The Balmer-continuum field is outside ATES’s XUV scope, so it is estimated from a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\infty} \frac{F_{\nu}}{h\nu} \sigma_2(\nu) d\nu, \quad F_{\nu} = \pi B_{\nu}(T_{\star}) \left(\frac{R_{\star}}{a}\right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu}\right)^3,$$

with ν_2 the $n = 2$ threshold and $\sigma_{2,\text{th}} \simeq 1.4 \times 10^{-17}$ cm², and $\Gamma_{2s} \simeq \Gamma_{2p} \simeq \Gamma_2$. This reproduces Huang et al.’s Table 2 values (~ 20 – 26 s⁻¹) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. ~ 140 s⁻¹ for HD 209458’s $T_{\star} \simeq 6065$ K), which suppresses n_2 and hence the Balmer-line depth. Setting $T_{\text{star}} \leq 0$ falls back to the manual constants `Gamma_2s/Gamma_2p` (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS machinery as Ly α . Along a chord at impact parameter b ,

$$\tau_\nu(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8 \text{ \AA}$ (air), $f_{23} = 0.6407$, $A_{3 \rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35 \text{ \AA}$ ($f_{24} = 0.119$, $A_{4 \rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, $\text{H}\alpha \simeq 3.2\%$ and $\text{H}\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_\star, T_\star$ (see the comparison notebook).

4. Implementation in TPM.py

- `n2_populations(T, n1s, ne, Jlya, G2s, G2p)` solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} . The Ly α -pump constants $B_{1s \rightarrow 2p}$, $B_{2p \rightarrow 1s}$ are precomputed.
- $J_{\text{Ly}\alpha}$ is read from `Jlya_file` if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile `data_n2` (in m^{-3}), exactly like `data_nHI`.
- A new wavelength grid `l_nde_Ha` and an H α LOS loop fill `exp_tau_Ha`; the disk average, instrument convolution (`Instr_res_Ha`) and rotation convolution produce the plotted curves.
- The H α branch always runs (`do_Ha=True`); with the default `Jlya_file=''` it uses the Huang et al. (2017) estimate for $J_{\text{Ly}\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).
- **Compatibility fix.** `Ion_species.txt` in ATES-metal carries trace-metal columns (16 columns total), so the reader was changed to `usecols=range(7)` to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs.

5. Validation

Run end-to-end on the converged HD 209458b ATES-metal output (`output/*_adv.txt`), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 $\text{\AA}$. With the Huang $J_{\text{Ly}\alpha}$ estimate ($\xi = 0.5$, $F_{\text{LyC}} \simeq 5.2 \times 10^2$, Γ_2 from $T_\star = 6065 \text{ K}$), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as

before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook `HalpHa_compare_HD209458b.ipynb`.

6. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter `T_star`); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting `T_star` ≤ 0 reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}}$ from ATES is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_ν at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

Reference

- Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144 — “H α Absorption in Transiting Exoplanet Atmospheres” (references/Christie_2013ApJ_772_144.pdf).
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150 — “A Model of the H α and Na Transmission Spectrum of HD 189733b” (references/Huang_2017_ApJ_851_150.pdf); the $J_{\text{Ly}\alpha}$ estimate is their Eq. (6) and related text.
- Jensen, A. G., et al. 2012, ApJ, 751, 86 — transit H α spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).